

Homework — 1.3.3 Networks — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [2] Row 1: **Application** layer; Row 2: **Transport** layer; Row 3: **Internet** (network) layer; Row 4: **Link** layer. (2 correct rows per mark.)

2. [4] 1 mark per correct row:

- Row 1: **True**.
- Row 2: **False** — packets are routed independently and different packets may take **different** routes to the destination.
- Row 3: **True**.
- Row 4: **False** — the internet is the largest example of a **WAN** (it spans a huge geographical area over third-party infrastructure).

3. [3] In the order the rows are printed: **3, 4, 1, 6, 2, 5**. Correct sequence: 1 — user types the URL and presses Enter; 2 — browser sends the domain name to a DNS server; 3 — DNS server resolves the domain name to an IP address and returns it; 4 — browser sends a request to the web server at that IP address; 5 — the response is split into packets that may be routed independently; 6 — packets are reassembled using sequence numbers and the page is displayed. (2 correct rows per mark.)

4. [3] Two advantages (1 each, max 2): security/user accounts/access levels are managed centrally; backups can be performed centrally; files and software are stored/updated in one place; easier to manage as the network grows. One disadvantage (1): the server is a single point of failure; expensive — dedicated server hardware and technical staff needed; the server can become a bottleneck under heavy demand. Must have two advantages and one disadvantage.

5. [2] The firewall monitors incoming and outgoing traffic (1) and blocks or permits it according to a set of rules (packet filtering), preventing unauthorised access to the network (1).

Part B (6 marks)

6. [2] Every foreign key value must match an existing primary key value in the table it references (1). This keeps links between tables consistent — e.g. a record cannot reference a record that does not exist, and a record cannot be deleted while other records still refer to it (prevents orphaned records) (1).

7. [2] Any two for 1 mark each: one-way / cannot be reversed to recover the input; deterministic — the same input always produces the same hash; low chance of collisions (different inputs rarely give the same hash); quick to calculate; produces a fixed-length output.

8. [2] The linker combines the compiled object code of the program with the object code of the library modules/subroutines it uses (1) and resolves the addresses/references between them so they form a single working executable (1).

Total: 14 + 6 = 20 marks